

“Models” subsumes “satisfies” (Wording for US298 and US300)

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§ 1 Introduction

This paper provides wording to resolve NB comments [US298](#) and [US300](#). It also resolves [[LWG3345](#)].

§ 2 Wording

This wording is relative to [[N4849](#)].

Add the following paragraphs to 16.5.4.12 [[res.on.requirements](#)] before p1:

- 1 A sequence *Args* of template arguments is said to *model* a concept *C* if *Args* satisfies *C* (13.5.2 [[temp.constr.decl](#)]) and meets all semantic requirements (if any) given in the specification of *C*.
- 2 If the validity or meaning of a program depends on whether a sequence of template arguments models a concept, and the concept is satisfied but not modeled, the program is ill-formed, no diagnostic required.

§ 3 References

[LWG3345] Daniel Krügler. Incorrect usages of “models” versus “satisfies”.
<https://wg21.link/lwg3345>

[N4849] Richard Smith. 2020. Working Draft, Standard for Programming Language C++.
<https://wg21.link/n4849>